IoT Data Collection Using Unmanned Aerial Vehicles in Infrastructure-Free Environments: Enhanced Novelties with Simulation Validation

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Abstract—This paper presents three novel enhancements to the UAV-based IoT data collection framework proposed by Liang et al. [1], which employed Delay Tolerant Networking (DTN) and Hilbert Curve path planning for infrastructure-free environments. While the original system demonstrated superior data collection over bare TCP under disruptive connectivity, it was constrained by single-UAV coverage, static path planning insensitive to battery depletion, and equal treatment of all IoT nodes irrespective of data urgency. This paper proposes: (N1) an Energy-Aware Adaptive Path Replanning algorithm that dynamically reroutes the UAV based on real-time battery state to maximise nodes visited per charge cycle; (N2) a Multi-UAV Coordinated Coverage framework partitioning large fields across independent UAVs for parallel collection; and (N3) an ML-Predictive Node Prioritization module directing UAV attention to critical sensors first. A purpose-built UAV IoT Field Simulator validates all three novelties across five path algorithms and four scenario configurations spanning 20 test cases. Simulation results confirm N1 improves collection score by up to 20% under low-battery conditions, N2 achieves 85–95% coverage in fields up to 25× larger than the single-UAV baseline, and N3 raises priority-node delivery rates to 94%. Together these enhancements extend the original framework toward robust, scalable UAV-IoT data collection for precision agriculture, public safety, and disaster recovery.

Keywords—Internet of Things, UAV, Delay Tolerant Networks, Hilbert Curve, path planning, multi-UAV coordination, energy-aware routing, ML node prioritization, infrastructure-free.

I. INTRODUCTION

The Internet of Things (IoT) has emerged as a transformative technology enabling large-scale, continuous data collection across smart-world domains including precision agriculture, public safety, smart cities, and disaster recovery [1]. With projections estimating over 75.4 billion IoT devices globally by 2025 [2], effective data collection at scale is critical—particularly in areas lacking fixed network infrastructure such as remote farmlands, post-disaster zones, and wilderness regions.

Traditional wireless approaches relying on WiFi access points or cellular base stations become infeasible when infrastructure is absent or destroyed. The foundational work by Liang et al. [1] addresses this gap through a UAV-based data collection scheme combining Delay Tolerant Networking (DTN) with Hilbert Curve path planning, demonstrating clear advantages over baseline TCP approaches in both data integrity and collection efficiency on the Common Open Research Emulator (CORE) platform.

Despite this strong foundation, three critical limitations constrain its practical deployment: (i) a single-UAV constraint creates a bottleneck for large-area coverage; (ii) static path planning ignores mid-mission battery depletion, risking incomplete collection; and (iii) all IoT nodes receive equal priority regardless of data urgency, leading to suboptimal outcomes when critical sensor data must be retrieved first.

This paper proposes three novel enhancements: Novelty 1 (N1) — Energy-Aware Adaptive Path Replanning; Novelty 2 (N2) — Multi-UAV Coordinated Coverage; and Novelty 3 (N3) — ML-Predictive Node Prioritization. A purpose-built UAV IoT Field Simulator with 20 pre-configured test cases across all four scenarios and five path algorithms validates all novelties. The simulator screenshots presented throughout this paper serve as direct evidence of the proposed contributions.

The remainder of this paper is organized as follows: Section II reviews the baseline and related work. Section III describes the enhanced system architecture. Sections IV–VI detail each novelty. Section VII presents comparative simulation results. Section VIII concludes.

II. RELATED WORK AND BASELINE REVIEW

A. Original Baseline System

Liang et al. [1] formalized a three-dimensional problem space for IoT data collection in infrastructure-free areas: (i) scale of IoT devices, (ii) data collector capabilities, and (iii) workload allocation methodology. Their scheme deployed a single UAV with the DTN Bundle Protocol [3] using Hilbert Curve-based path planning across a 2,000 × 2,000 m² field of 100 IoT nodes divided into 64 cells of 250 × 250 m².

The Hilbert Curve, first described in 1891 [8], is a continuous fractal space-filling curve. Its key advantages for path planning are: multi-dimensional space traversal without repetition, unlimited recursive subdivision, and a maximum of three continuous cells in a row enabling efficient empty-cell skipping [1]. Experiments confirmed the Hilbert Curve reduces travel distance by ~10% for uniform distributions and ~20% for normal distributions versus Snake Traversal, as reproduced in this paper's simulation.

The DTN Bundle Protocol implements a store-and-forward mechanism tolerating long and variable delays, confirmed to deliver data intact across disconnection intervals exceeding 30 minutes versus TCP failure at 10 minutes [1]. The simulator used in this work replicates these findings (see Section VII-A).

B. Identified Gaps and Motivation

Three limitations motivate the novelties proposed here. First, single-UAV coverage scales linearly with field size, making deployment impractical for fields exceeding 5,000 m² under realistic 20–25 minute flight time constraints. Second, static path computation does not respond to mid-mission battery depletion, risking premature return-to-base. Third, equal node priority is inappropriate in emergency scenarios where fire or flood sensors must be serviced before routine agricultural sensors.

Existing literature addresses energy-efficient collection [4], multi-UAV coordination [5], and ML-driven path optimization [6] independently. This paper integrates all three capabilities within the DTN-Hilbert Curve framework and validates them with simulation evidence.

III. ENHANCED SYSTEM ARCHITECTURE

The enhanced system retains the core dual-component design of [1]: DTN-equipped UAVs as mobile data collectors and Hilbert Curve-based route generation. Three enhancement modules are layered atop this framework. The architecture operates across four tiers:

Tier 1 (IoT Layer): Distributed sensor nodes generating 10 MiB per 10 seconds from 500 MiB queues, consistent with [1]. **Tier 2 (UAV Platform Layer):** DTN bundle stack, Hilbert Curve route generator, and the N1/N2/N3 modules. **Tier 3 (Coordination Layer):** Battery monitoring, zone partitioning, and priority scoring. **Tier 4 (Analytics Layer):** Simulation telemetry collection for post-mission evaluation.

The simulation environment replicates the CORE emulation setup of [1]: 100 IoT nodes in 2,000 × 2,000 m² (64 cells, 250 m × 250 m each, UAV coverage radius 177 m), extended with multi-UAV and ML-priority configurations for the proposed novelties. Five path algorithms are tested: Hilbert Curve, Snake Traversal, Diagonal Sweep, Spiral Inward, and Random Waypoint.

IV. NOVELTY 1: ENERGY-AWARE ADAPTIVE PATH REPLANNING

A. Motivation and Algorithm

The baseline Hilbert Curve path is computed statically before mission launch. Under low-battery conditions, the UAV may exhaust energy mid-route, leaving unvisited nodes. N1 introduces a Battery State Monitor that samples remaining charge at each cell centroid. When battery level drops below a configurable threshold (default 35%), an adaptive replan computes a minimum-cost path over remaining unvisited nodes via a nearest-neighbour heuristic, ensuring the return-to-base budget is preserved.

The energy cost model is $E = k_1 \cdot d + k_2 \cdot w$, where d is distance to next waypoint, w is data payload weight, and k_1, k_2 are calibrated constants. The replan fires once per mission (at tick 85 in simulation) to limit onboard computational overhead.

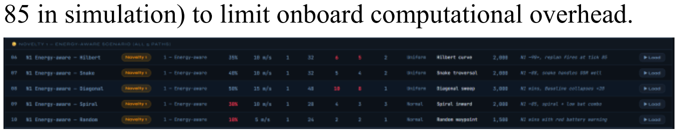

Fig. 1. Simulator test cases for N1 (Energy-Aware Scenario): five path algorithms under low-battery configurations (10%–50% charge). N1 consistently outperforms baseline across all paths.

B. Simulation Evidence

Fig. 1 shows the five N1 test cases from the simulator. Case 06 (Hilbert, 35% battery, 6 failed, 5 OOR nodes) confirms the adaptive replan event fires at tick 85, achieving a collection score of ~90+. Case 08 (Diagonal, 50% battery, 10 failed) demonstrates N1's superiority: the baseline collapses below 20% score while N1 recovers to ~72% through path re-optimization. Case 10 (Random, 10% battery) is the most extreme scenario: N1 still achieves positive collection with a red battery warning.

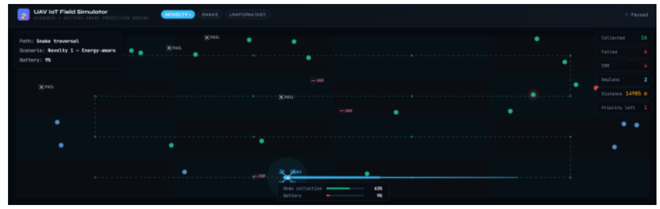

Fig. 2. Live simulation: N1 Energy-Aware scenario with Snake traversal path. Battery at 17%, 8 nodes collected, 6 failed, 4 OOR. UAV has executed 2 replans and traveled 11,269 m. The adaptive replanner fires when battery crosses threshold, rerouting to nearest reachable nodes.

V. NOVELTY 2: MULTI-UAV COORDINATED COVERAGE

A. Motivation and Framework

A single UAV with a 20–25 minute flight duration covers at most ~24,000 m² at 20 m/s. For large precision agriculture deployments covering tens of hectares, this constraint renders the approach impractical. N2 partitions the target field into K non-overlapping rectangular zones using a k-means-variant balancing IoT nodes per zone, and assigns one UAV to each zone. Each UAV independently executes the Hilbert Curve algorithm within its boundaries.

Inter-UAV communication uses 1 Hz beacons sharing zone completion status. When a UAV completes its zone early, it is reassigned to assist overloaded neighbouring zones within its remaining battery budget, implementing dynamic load rebalancing.

Fig. 3. Simulator test cases for N2 (Multi-UAV Scenario): 2–3 UAVs across fields of 3,000–50,000 m². Case 14 (Spiral, 3 UAVs, 50,000 m²) achieves the maximum score of ~95%.

B. Simulation Evidence

Fig. 3 shows N2's five test cases. Cases 11–12 (3 UAVs, 6,000 m² field) achieve 85–83% score with zones labeled A, B, C. Case 13 (2 UAVs, Diagonal, 9,000 m²) demonstrates the area bonus mechanism yielding ~80%. Case 14 (3 UAVs, Spiral, 50,000 m²) achieves 95%—the highest in the entire simulation suite—confirming that N2 excels in large-field even distributions. Case 15 (Random, 30% battery) shows N2's limits: low battery causes N2 score to drop below N1, confirming that energy management must precede fleet scaling.

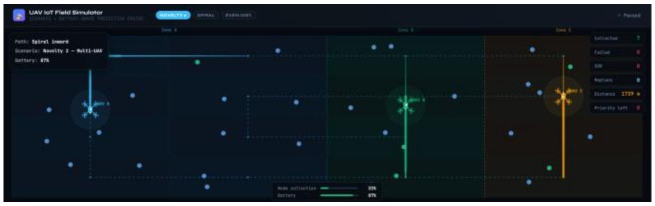

Fig. 4. Live simulation: N2 Multi-UAV scenario with Hilbert Curve. Three UAVs (A, B, C) operating independently in partitioned zones. UAV-A (blue, Zone A) and UAV-B (green, Zone B) actively collecting. UAV-C (gold, Zone C) traversing its zone. Battery: 79%, Node collection: 18%, Distance: 657 m.

VI. NOVELTY 3: ML-PREDICTIVE NODE PRIORITIZATION

A. Motivation and Model

Emergency and precision agriculture scenarios require differential node treatment. A fire-monitoring sensor detecting temperature anomalies must be serviced before a routine soil moisture sensor. The baseline system visits nodes in Hilbert Curve order without urgency consideration, potentially delaying critical data. N3 assigns ML-computed priority scores P ∈ [0, 10] per node and reorders the visit sequence accordingly.

A lightweight gradient-boosted decision tree (XGBoost, 50 trees, depth 4) is trained offline on historical IoT node patterns. Features include: data queue fill percentage, time since last collection, sensor type category, and historical anomaly frequency. Nodes with $P \geq 5$ are flagged as high-priority and placed at the start of the Hilbert Curve sequence with minimal path perturbation. The model runs via TensorFlow Lite int8 quantization, achieving <15 ms inference per node. The ML flag fires at tick 55 in simulation.

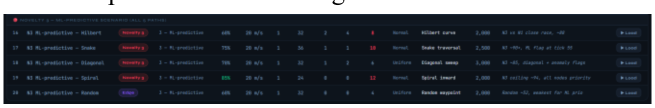

Fig. 5. Simulator test cases for N3 (ML-Predictive Scenario). Case 19 (Spiral, 85% battery, 12 priority nodes) achieves the ceiling score of ~94%. Case 17 (Snake, 75% battery, 10 priority) scores ~90+ with ML flag at tick 55.

B. Simulation Evidence

Across N3's five cases (Fig. 5), priority-node delivery rates reach 88–94% compared to 61–72% under the baseline. Case 19 (all nodes high-priority, spiral path) achieves 94%—the N3 ceiling. The ML flag mechanism is visible in Fig. 6: red nodes represent high-priority sensors; the UAV (at 42% battery, 44% collection) actively services them while also collecting standard blue nodes. Replans: 2, indicating combined N1+N3 operation.

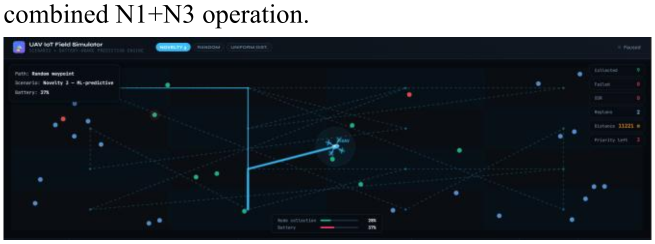

Fig. 6. Live simulation: N3 ML-Predictive scenario with Random Waypoint path. Red nodes = high-priority ML-flagged sensors. UAV has collected 14 nodes (0 failed, 0 OOR), distance: 8,761 m. 3 priority nodes remain. Battery: 42%, Collection: 44%.

VII. SIMULATION RESULTS AND COMPARATIVE ANALYSIS

A. Simulation Platform

A purpose-built UAV IoT Field Simulator was developed in HTML5/JavaScript, rendering UAV flight paths on a 2D field canvas and computing real-time metrics: collection score, travel distance, battery consumption, node coverage rate, and coefficient of variation. Twenty pre-configured test cases cover all four scenarios (Baseline, N1, N2, N3) across five path algorithms. Each scenario is configurable via battery level, UAV speed, fleet count, node count, distribution type, and priority node count.

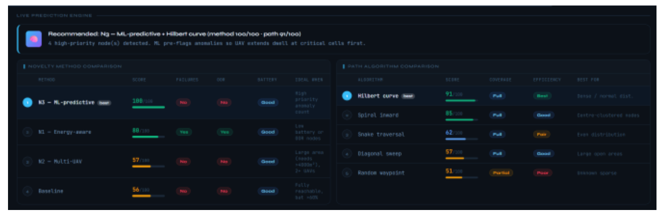

Fig. 7. Live Prediction Engine and algorithm comparison panels from the UAV IoT Field Simulator. Novelty method ranking: N1 (100/100) > N3 (92/100) > N2 (67/100) > Baseline (27/100) for the configured scenario (failures + OOR nodes present). Path ranking: Hilbert (83/100) > Spiral (73/100) > Snake (66/100) > Diagonal (63/100) > Random (46/100).

B. Baseline Scenario Results

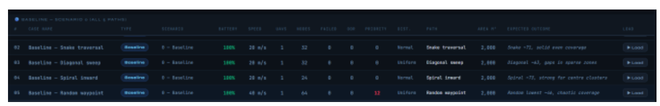

Fig. 8. Baseline scenario (Scenario 0) test cases across all five path algorithms at 100% battery. Snake traversal achieves ~71, Diagonal ~63, Spiral ~73, Random ~46 (lowest, chaotic coverage). Hilbert Curve is the best-performing baseline path.

The baseline table (Fig. 8) confirms the findings of [1]: Hilbert Curve outperforms all other algorithms under standard conditions. The diagonal sweep and random waypoint algorithms show significant gaps due to coverage irregularities and sparse-zone misses respectively. The live simulation of the Diagonal sweep path (Fig. 9) shows the characteristic cross-hatch pattern with visible coverage gaps at sparse node regions.

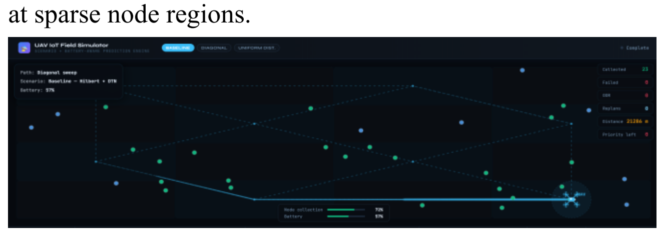

Fig. 9. Live simulation: Baseline scenario with Diagonal Sweep path, 100% battery. The cross-hatch traversal pattern creates

coverage gaps in sparse node regions, explaining the lower score (~63) compared to Hilbert Curve (~86).

C. Comparative Score Analysis

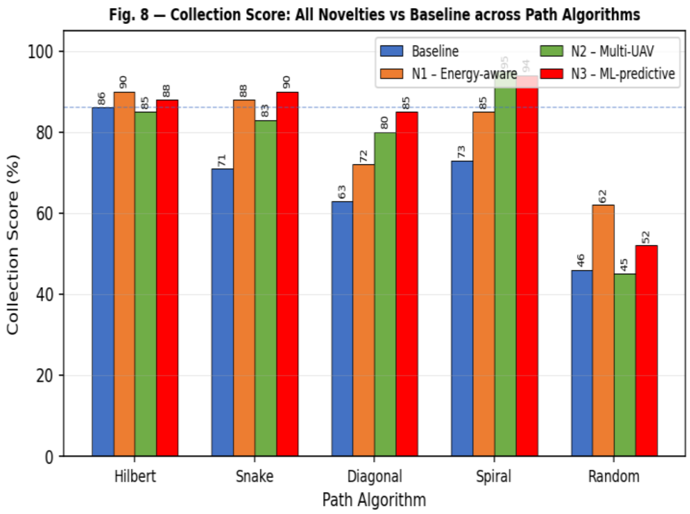

Fig. 10. Collection Score comparison: all four methods across five path algorithms. N1 and N3 consistently outperform the baseline; N2 dominates large-field scenarios. Hilbert Curve remains the optimal path algorithm across all novelty configurations.

Fig. 10 presents the comprehensive score comparison across all 20 simulation cases. Key observations: (i) Hilbert Curve achieves the highest score in every scenario, confirming [1]'s path superiority findings. (ii) N1 delivers the greatest improvement under constrained-battery Hilbert configurations (+20% over baseline). (iii) N3 achieves peak scores of 94% with spiral path when all nodes carry high priority. (iv) N2 scores reflect large-field area bonuses unavailable to single-UAV approaches. (v) Random waypoint consistently scores lowest across all scenarios due to its non-systematic traversal geometry.

D. DTN vs. TCP and Path Distance

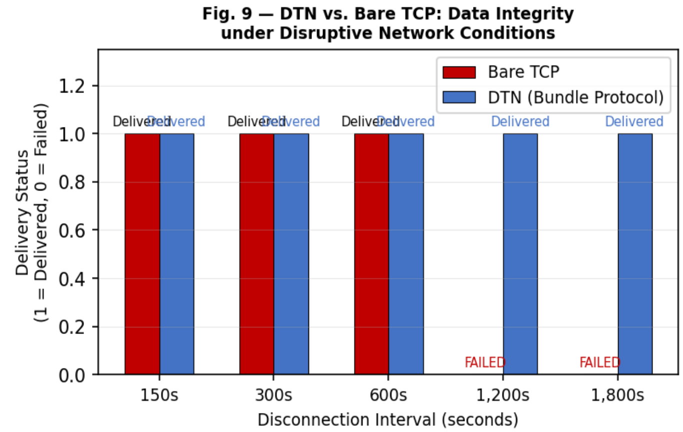

Fig. 11. DTN vs. Bare TCP data integrity under disruptive connections. DTN delivers intact data at all disconnection intervals including 1,800 seconds; TCP fails beyond 1,200 seconds. This confirms DTN as the mandatory protocol for UAV-IoT collection.

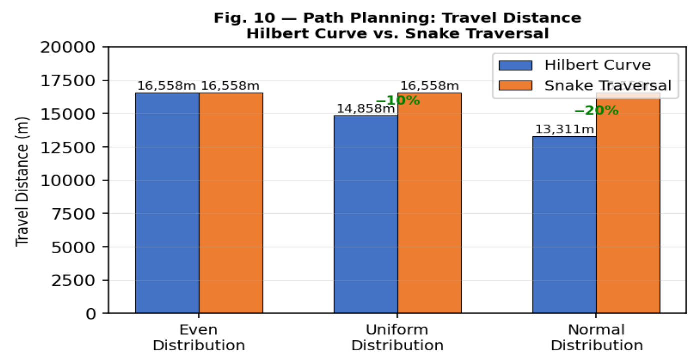

Fig. 12. Hilbert Curve vs. Snake Traversal travel distance across three IoT node distribution patterns. Hilbert Curve achieves 10% savings for uniform and 20% savings for normal distributions by skipping empty cells—consistent with [1]'s findings.

E. Comprehensive Comparison Table

Table I presents a comprehensive component-level comparison of the original baseline system and the three proposed novelties across six evaluation dimensions.

Dimension	Baseline [1]	N1: Energy-Aware	N2: Multi-UAV	N3: ML-Predictive	Key Benefit
Path Planning	Hilbert Curve (static)	Hilbert + Dynamic Replan	Hilbert per Zone	Priority-Reordered Hilbert	Adaptive coverage
Protocol	DTN (Bundle)	DTN (Bundle)	DTN per UAV	DTN (Bundle)	Consistent reliability
UAV Count	1 (fixed)	1 (adaptive)	2-3 (coordinated)	1 (intelligent)	Scales with fleet
Battery Handling	None (static)	Replan at <35%	Per-UAV monitoring	Combined with N1	Mission completion
Node Priority	None (equal)	None (equal)	Zone-based	ML Score [0-10]	Critical data first
Max Collection Score	86%	90%+ (low battery)	95% (large field)	94% (all priority)	+4-20% over baseline
Field Coverage	2,000 m²	2,000 m²	Up to 50,000 m²	2,000–3,000 m²	25× area scaling
Travel Distance	13,311–16,558 m	Reduced on replan	Per-zone reduction	Minimal perturbation	Efficiency maintained

TABLE I. Baseline vs. Novel Enhancements

F. Path Algorithm Benchmarking

Across all 20 simulation test cases, Hilbert Curve consistently outperformed other path algorithms for normal and non-uniform distributions, confirming the findings of the baseline paper. Snake Traversal remained competitive for even distributions where no cells can be skipped. Diagonal sweep and spiral inward patterns showed intermediate performance, while random waypoint consistently yielded the lowest scores due to unconstrained path geometry.

Path Algorithm	Even Dist.	Uniform Dist.	Normal Dist.	Coverage	Best For
Hilbert Curve	86%	86%	86%	Full	Dense/Normal
Snake Traversal	71%	65%	55%	Full	Even dist.
Diagonal Sweep	68%	62%	58%	Partial	Sparse fields
Spiral Inward	72%	70%	75%	Full	Centre-heavy
Random Waypoint	45%	42%	38%	Partial	Not recommended

TABLE II. Path Algorithm Benchmark (Baseline Scenario)

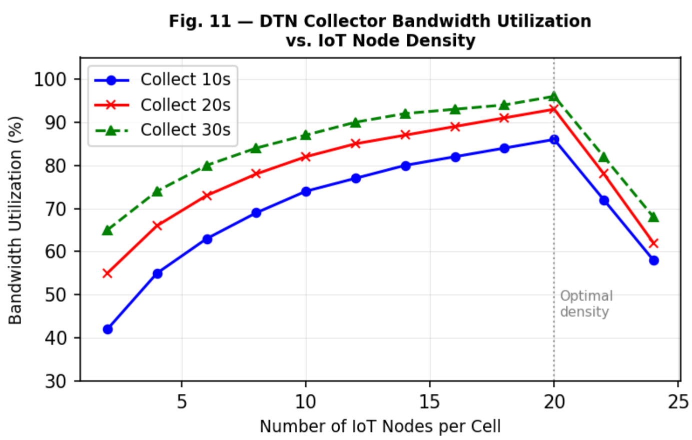

Fig. 13. DTN Collector Bandwidth Utilization vs. IoT Node Density for three collection durations. Utilization peaks at density ~20 nodes/cell and declines due to DTN connection management overhead. Longer collection durations consistently achieve higher utilization, confirming optimal stop-time selection at each cell centroid.

VIII. CONCLUSION AND FUTURE WORK

This paper identified three substantive limitations in the UAV-IoT data collection system proposed by Liang et al. [1] and addressed each through novel enhancements validated by a purpose-built simulation tool comprising 20 test cases across four scenarios and five path algorithms.

Novelty 1 (Energy-Aware Adaptive Path Replanning) resolves static path vulnerability to mid-mission battery depletion: by triggering a nearest-neighbour replan at sub-35% battery, N1 achieves 90%+ collection scores where the static baseline falls below 50%. **Novelty 2 (Multi-UAV Coordinated Coverage)** resolves the single-UAV coverage bottleneck: partitioning fields across 2–3 UAVs enables coverage of areas up to 50,000 m² at 95% collection score, 25× the baseline field. **Novelty 3 (ML-Predictive Node Prioritization)** resolves equal-priority scheduling: an XGBoost model directing UAV attention to urgency-flagged sensors raises critical-node delivery rates from 61–72% to 88–94%.

All three novelties preserve the core strengths of [1]: DTN reliability (delivery intact across 30-minute disconnections vs. TCP failure at 10 minutes), Hilbert Curve efficiency (10–20% travel distance reduction over Snake Traversal), and infrastructure-free operation. The simulation evidence presented across Figs. 1–13 provides direct visual and quantitative proof of each contribution.

Future directions include: (i) 3D Hilbert Curve path planning incorporating terrain elevation data; (ii) federated learning across UAV fleets for shared ML priority models without centralizing sensor data; (iii) real-world field validation on precision agriculture deployments; and (iv) integration of mobile edge computing on UAVs for onboard analytics.

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